

Total No. of Questions : 4]

SEAT No. :

PF4

[Total No. of Pages : 2

APR-26/FE/Insem-4
F.E. (All Branches) (Insem)
BASIC ELECTRICAL ENGINEERING
(2019 Pattern) (Semester-II) (103004) (Credit System)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2 and Q.3 or Q.4.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable additional data, if necessary.*
- 5) *Use of non-programmable calculator is allowed.*

- Q1)** a) Compare magnetic & electric circuit for similar & dissimilar points. [7]
b) A coil of N turns is uniformly wound on a cast iron ring having mean length of 50 cm and area of cross section 12.56 cm^2 . A current of 2 A in the coil, produces flux of 6 mWb in the air gap of 2 mm length. The relative Permeability of the iron is 1000. Calculate: [8]

- i) Reluctance of iron path
- ii) Reluctance of air gap and
- iii) The number of turns N .

OR

- Q2)** a) State and explain Faradays laws of Electromagnetic induction. [7]
b) Define the following terms. [8]
Magnetic flux density, permeability, relative permeability, reluctance, mmf, self & mutual inductance and coefficient of coupling

- Q3)** a) Define the average value and obtain the expression for average value of sinusoidally varying alternating current in terms of its peak value. [7]
b) Three capacitors of $2 \mu\text{F}$, $3 \mu\text{F}$ and $6 \mu\text{F}$ respectively are connected in series across a 500 V d.c. supply. Calculate: [8]
i) Total capacitance
ii) The charge on each capacitor
iii) The p.d. across each capacitor; and
iv) The energy stored in the $6 \mu\text{F}$ capacitor.

OR

P.T.O.

- Q4)** a) Derive an expression for Energy stored in Capacitor. [7]
- b) A sinusoidally varying alternating voltage of 100V with 50 Hz frequency is applied to a circuit. find: [8]
- i) The mathematical equation of the voltage
 - ii) Time Period
 - iii) The instantaneous voltage when $t = 0.3 \text{ ms}$
 - iv) The time taken by voltage to reach 85V
 - v) Average value of the voltage
 - vi) Maximum value of the voltage.

